

B Simple Interest & Compound Interest - 1

14/8/25

* Simple Interest (SI) :

Sol: $R = 10\%$ / Annum.

Investment

$$P = ₹ 100 \xrightarrow{1 \text{ year}} ₹ 110 = 10\%$$

money
sum

$$P = ₹ 100 \xrightarrow{10\%} ₹ 120$$

only 10% for 100

$$P = ₹ 100 \xrightarrow{10\%} ₹ 130$$

total

Amount = 130
bringing back.

Formula

$$SI = \frac{PRT}{100}$$

P = principle

R = Rate of Interest

T = Time 1 year or etc.

Sol: $T \rightarrow 1 \text{ yr}$

$T \rightarrow 6 \text{ m} = \frac{6}{12} \text{ yr}$

$T \rightarrow 1 \text{ m} = \frac{1}{12} \text{ yr}$

Amount

$$A = P + I$$

$$A = 100 + 30$$

P. ₹.

* Compound Interest :-

$R = 10\% / \text{yr}$

$$P_1 = ₹ 100 \xrightarrow{10\% \rightarrow 10} 110$$

$$P_2 = ₹ 110 \xrightarrow{10\% \rightarrow 11} 121$$

$$P_3 = ₹ 121 \xrightarrow{10\% \rightarrow 12.1} 133.1$$

$$CI = P \left(1 + \frac{r}{100} \right)^n - P$$

n $\rightarrow$ Time

r $\rightarrow$ Rate of Interest

P $\rightarrow$ Principle

$$A = P + I$$

$$A = P \left(1 + \frac{r}{100} \right)^n$$

$$A - P = I$$

$$CI = A - P$$

A. $\rightarrow$ 1st Big $\uparrow$

P $\Rightarrow$ 2nd Big T

I $\rightarrow$ 3rd Big T

B3 Simple Interest Questions :-

① What would be the Simple Interest obtained on an amount of Rs. 1200 at the rate of 10% per annum after 6 years?

Solⁿ

W.K.T Given:-

$$P = 1200$$

$$R = 10\%$$

$$T = 6 \text{ yrs.}$$

W.K.T

$$SI = \frac{PRT}{100} = \frac{1200 \times 10 \times 6}{100} = 720$$

Ans.

② What would be the Simple Interest obtained on an amount of Rs. 6535 at the rate of 10% per annum after 6 years.

$$SI = \frac{PRT}{100} = \frac{6535 \times 10 \times 6}{100} = 3921$$

Ans

③ What would be the Simple Interest obtained on an amount of Rs. 5760 at the rate of 6% per annum after 3 years?

$$SI = \frac{PRT}{100} = \frac{5760 \times 6 \times 3}{100} = 1036.8$$

$$= 1036.8$$

Ans

4) How much time will it take for an amount of Rs. 450 to yield Rs. 81 as interest at 4.5% per annum of simple interest.

Sol

$$SI = \frac{PRT}{100}$$

$$S.I = 81$$

$$P = 450$$

$$R = 4.5\%$$

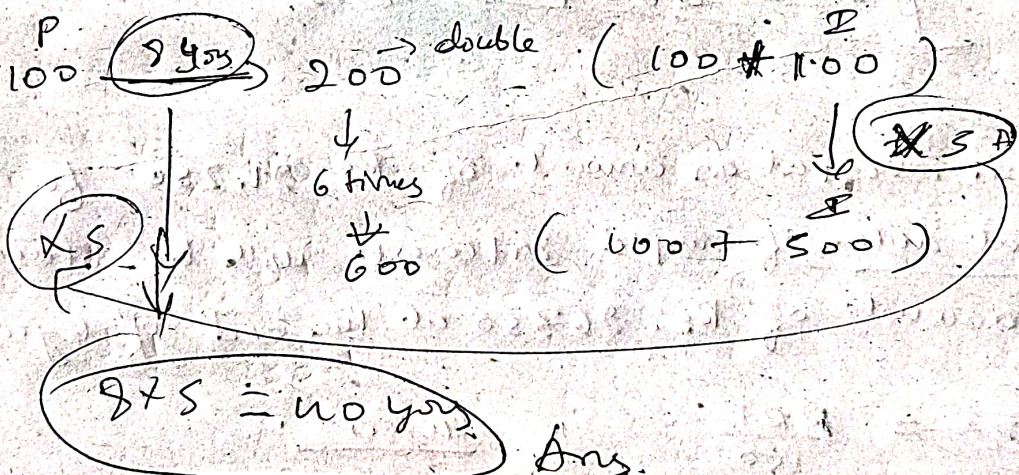
$$T = \frac{SI \times 100}{R \times P} = \frac{81 \times 100}{4.5 \times 450} = \frac{48}{4.5} = 4$$

$$T = 4 \text{ yrs}$$

Ans

5) A sum of money doubles itself in 8 years. In how many years will it be 6 times?

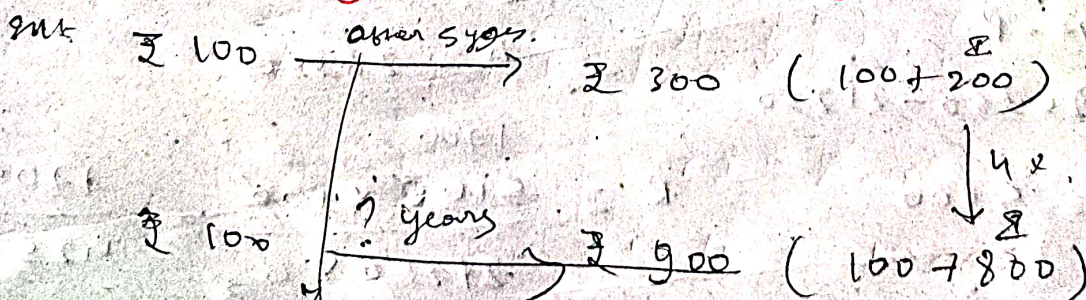
Sol



Ans.

Sol

6) A sum of money triples itself in 5 years. In how many years will it be 9 times?



$$3 \times 5 = 15 \text{ yrs}$$

Ans

7) Mr. Raghav invested a sum of Rs. 15,400 at simple interest for 6 years at 10%. After 6 years, Mr. Raghav got an amount of Rs. 24,704. What is the simple interest he got after 6 years?

$$15400 \xrightarrow[6 \text{ years}]{10\%} 24,704$$

(P)

(A)

$I = ?$

w.k.t

$$A = P + I$$

$$I = A - P$$

$$I = 24,704 - 15,400$$

$$I = 9,264 \text{ Ans}$$

8) Mr. Akash invested an amount of Rs 21,250 for 6 yrs at what rate of simple interest will he obtain the total amount of Rs. 26,350 at the end of 6 years?

w.k.t $SI = \frac{PRT}{100}$

$$R = \frac{SI \times 100}{PT}$$

$$SI = A - P$$

$$= 26,350 - 21,250$$

$$SI = 5,100$$

Given :-

$$P = 21,250$$

$$T = 6 \text{ yrs}$$

$$SI = ?$$

$$R = ?$$

$$A = 26,350$$

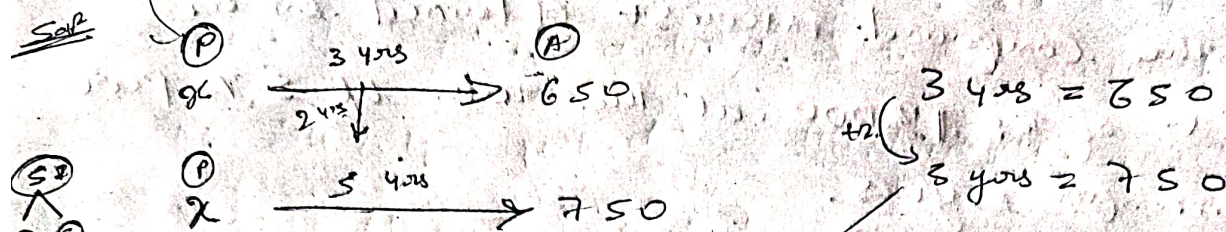
$$R = \frac{5100 \times 100}{21250 \times 6} = \frac{1700}{425}$$

$$R = 4\% \text{ Ans}$$

Ans

but not in opt

9) A sum of money at simple interest amounts to Rs. 650 in 3 years & Rs. 750 in 5 years. What is the principal?



$$2 \text{ yrs} = 100$$

$$\text{Interest (1 yr)} = \frac{100}{2} = 50$$

3 yrs = 650 (Amount (A))

3 yrs = 650 total (Principal (P) + Interest (I))

3 yrs × 50 = 150 (only interest)

∴ W.K.T

$$A = P + I$$

$$P = A - I$$

$$P = 650 - 150$$

$$P = 500 \text{ Ans}$$

10) Shashi took a loan of Rs. 1200 with simple interest for as many years as the rate of interest. If he paid Rs. 432 as interest at the end of the loan period, what was the rate of interest?

Given $P = ₹ 1200$

$$R = \frac{SI \times 100}{P \times T}$$

$$T = R$$

$$SI = ₹ 432$$

$$R = \frac{432 \times 100}{1200 \times R}$$

$$R^2 = \frac{432 \times 100}{1200}$$

$$R^2 = 36$$

$$R = \sqrt{36}$$

$$R = 6\% \text{ Ans}$$

Compound Interest Questions

- 11) Find the Compound Interest obtained on an amount of Rs 12000 at the rate of 5% per annum for 2 years.

$$\begin{array}{l}
 12000 \xrightarrow[5\%]{\rightarrow 1 \text{ year}} 12000 + 600 = 12600 \\
 12600 \xrightarrow[5\%]{\rightarrow 2 \text{ years}} 12600 + 630 = 13230
 \end{array}$$

$$\text{Total CI} = 1230$$

∴ C) Rs 1230 Ans.

formulae

$$CI = 2a + b$$

$$a = \frac{P \times r}{100}$$

$$b = \frac{a \times r}{100}$$

$$CI = 2 \left(\frac{12000 \times 5}{100} \right) + \left(\frac{12000 \times 5}{100} \right) \times \frac{5}{100}$$

$$= 2 \times 600 + \frac{3000 \times 5}{100}$$

$$= 2 \times 600 + 3000 \times \frac{5}{100}$$

$$= 1200 + 300$$

$$= \underline{1500}$$

Ans.

12) Find the C.I. obtained on an amount of Rs. 10,000 at the rate of 7% per annum for 2 yrs

$$10000 \xrightarrow{7\%} 10,700$$

$$10,700 \xrightarrow{7\%} 749 + 10,700 \rightarrow 749$$

$$1449$$

Ans

1449

$$C.I. = 2a + b$$

$$a = \frac{PR}{100}$$

$$b = a \left(\frac{R}{100} \right)$$

13) What would be the C.I. obtained on an amount of Rs. 4000 at the rate of 5% per annum after 3 yrs.

$$4000 \xrightarrow{5\%} = 40 \times 5 = 200 \rightarrow$$

$$4000 + 200 = 4200 \xrightarrow{5\%} = 210 + 4200$$

$$4410 \xrightarrow{\frac{5}{100} \times 4410} \rightarrow$$

$$\frac{5}{100} \times 4410 = 220.5$$

$$4410 + 220.5$$

$$4630.5$$

$$C.I._3 = 3a + 3b + c$$

$$a = \frac{PR}{100}$$

$$b = a \left(\frac{R}{100} \right)$$

$$c = b \left(\frac{R}{100} \right)$$

14) What would be the compound interest obtained on an amount of Rs. 5000 at the rate of 6% per annum after 3 yrs?

$$\begin{array}{l}
 5000 \xrightarrow{6\%} 5000 + 300 \\
 5300 \xrightarrow{6\%} 5300 + 318 \\
 5618 \xrightarrow{6\%} 5618 + 336.08 \\
 \hline
 \text{Total CI} = 955.08
 \end{array}$$

Ans: Not these

15) The difference b/w simple & compound interest compounded annually on a certain sum of money for 2 yrs at 4% per annum is Rs 1. The sum is...

$$T = 2 \text{ yrs}$$

Comp - Annually

Diff: b/w SI & CI

$$(SI - CI)_2 = P \left(\frac{R}{100} \right)^2$$

$$1 = P \left(\frac{4^2}{100} \right)^2$$

$$1 = P \left(\frac{4}{2500} \right)$$

$$\frac{2500}{4} = P$$

$$P = 625$$

Ans